

Functional Components of a Computer

1. What Are Functional Components?

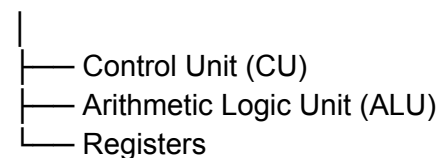
A computer performs its job through several major functional units.

The main ones are:

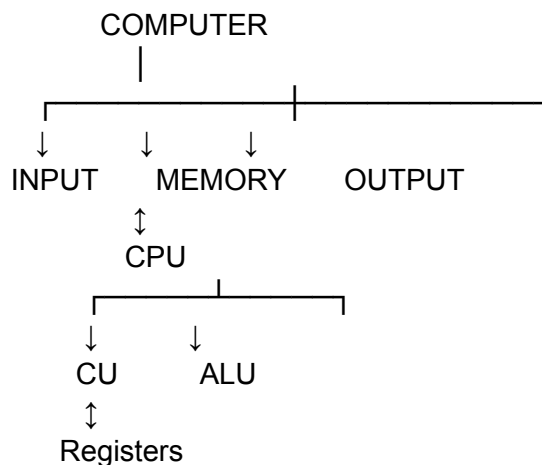
1. **Input Unit**
2. **Memory Unit**
3. **Central Processing Unit (CPU)**
4. **Output Unit**

The CPU itself contains important components:

CPU



So the overall picture is:



2. Input Unit

Let's start from the beginning.

What is Input?

Input is the data and instructions supplied to a computer.

Examples:

Keyboard → typing text

Mouse → clicking

Microphone → sound

Scanner → document/image

Camera → image/video

These devices allow information to enter the computer.

What does the Input Unit do?

The **Input Unit**:

Accepts data and instructions from the user or another source and sends them into the computer for processing.

For example:

You type:

25

+

10

↓

Keyboard

↓

Input Unit

↓

Computer

The input unit acts as the **entry point** for data.

3. Memory Unit

Once data enters the computer, it often needs to be stored so that the computer can use it.

That's the job of the **memory unit**.

Definition

The memory unit stores data, instructions, and intermediate/final results needed by the computer.

For example, when a program is running, the computer may need to keep:

Program instructions

Input data

Intermediate results

Temporary values

in memory.

4. Why Does the Computer Need Memory?

Imagine you're solving:

$$25 \times 10 + 50$$

The computer can't simply throw away every value immediately.

It needs to keep values while processing.

Conceptually:

$$25 \times 10$$

↓

$$250$$

↓

$$250 + 50$$

↓

$$300$$

Memory can hold the information required during this process.

5. CPU — The Processing Center

Now we reach the most important component.

Central Processing Unit

CPU = Central Processing Unit

It is the primary component responsible for executing instructions and performing processing operations.

A simple way to understand it:

The CPU executes program instructions and performs the operations required to process data.

Think of it as the computer's main **processing center**.

6. Output Unit

After processing, the computer needs to communicate the result to us.

That's the job of the **Output Unit**.

Definition

The output unit presents the processed results to the user or another system.

Examples:

Monitor → displays results

Printer → prints results

Speakers → produce sound

For example:

Input:

25 + 10

↓

Processing:

CPU calculates

↓

Output:

35

7. The Complete Functional Process

Now connect everything.

Suppose you use a calculator program and enter:

25 + 10

Step 1 — Input

You type the numbers using the keyboard.

Keyboard



Input Unit

Step 2 — Memory

The required data and instructions are held in memory.

Input → Memory

Step 3 — Processing

The CPU executes the required instructions.

Memory



CPU

Step 4 — Result

The calculation produces:

35

Step 5 — Output

The monitor displays:

35

So:

INPUT



MEMORY



CPU / PROCESSING



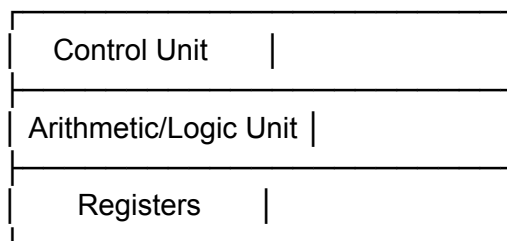
OUTPUT

8. But What Happens Inside the CPU?

This is where your second diagram becomes important.

Your slide shows:

CENTRAL PROCESSING UNIT



So CPU has three important parts in this simplified model:

1. Control Unit — CU

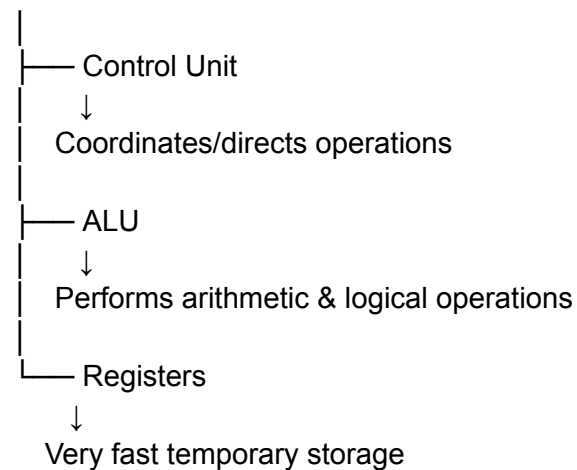
2. Arithmetic and Logic Unit — ALU

3. Registers

We'll study these in much more detail in **#6 CPU**.

For now, just understand their roles:

CPU



9. ALU

ALU = Arithmetic Logic Unit

As the name suggests, it performs two major categories of operations.

Arithmetic

For example:

+
-
×
÷

Logic

For example:

>
<
=
AND
OR
NOT

Suppose:

25 + 10

The arithmetic operation is performed by the **ALU**.

Suppose the computer needs to determine:

25 > 10

The ALU can perform the comparison.

10. Control Unit

The **Control Unit (CU)** coordinates the operations of the computer.

It essentially directs the different components:

"Get this instruction."

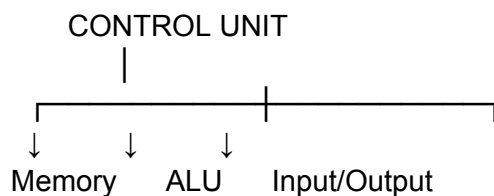
"Get this data."

"Perform this operation."

"Store the result."

"Send the result to the output."

A simplified view:

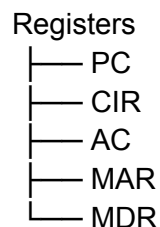


The CU doesn't primarily perform arithmetic itself.

Instead, it **controls and coordinates** the execution of instructions.

11. Registers

Your teacher's CPU diagram shows:



These are very small, extremely fast storage locations inside the CPU.

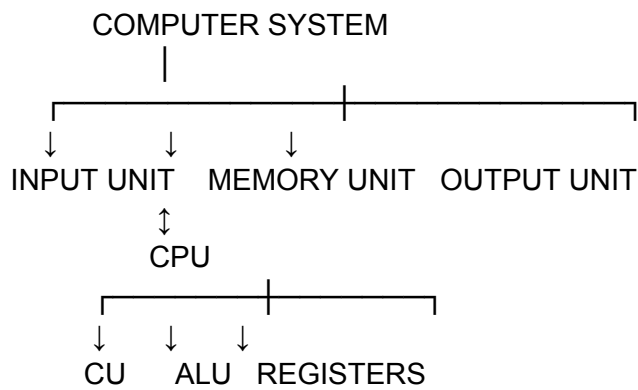
For now, remember:

Registers temporarily hold data, instructions, addresses, and intermediate results that the CPU needs immediately.

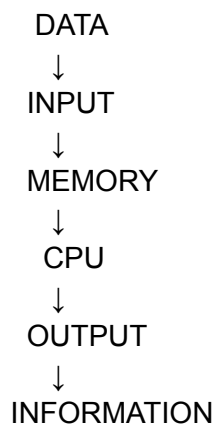
You will learn each register in detail when we study the **CPU**.

12. The Big Picture

Now look at the entire computer as one system:



And the basic flow:



★ One Important Correction

Your slide labels the memory/storage section as "**Memory Unit**" and also uses the word "**Storage.**"

In modern computer terminology, we distinguish:

Primary memory

Examples:

RAM
Cache
Registers

Used directly during processing.

Secondary storage

Examples:

SSD
HDD
USB drive

Used for longer-term storage.

So don't assume:

Memory = HDD/SSD

They are related but technically different concepts.

We'll deal with this distinction later when we study memory/storage in more depth.